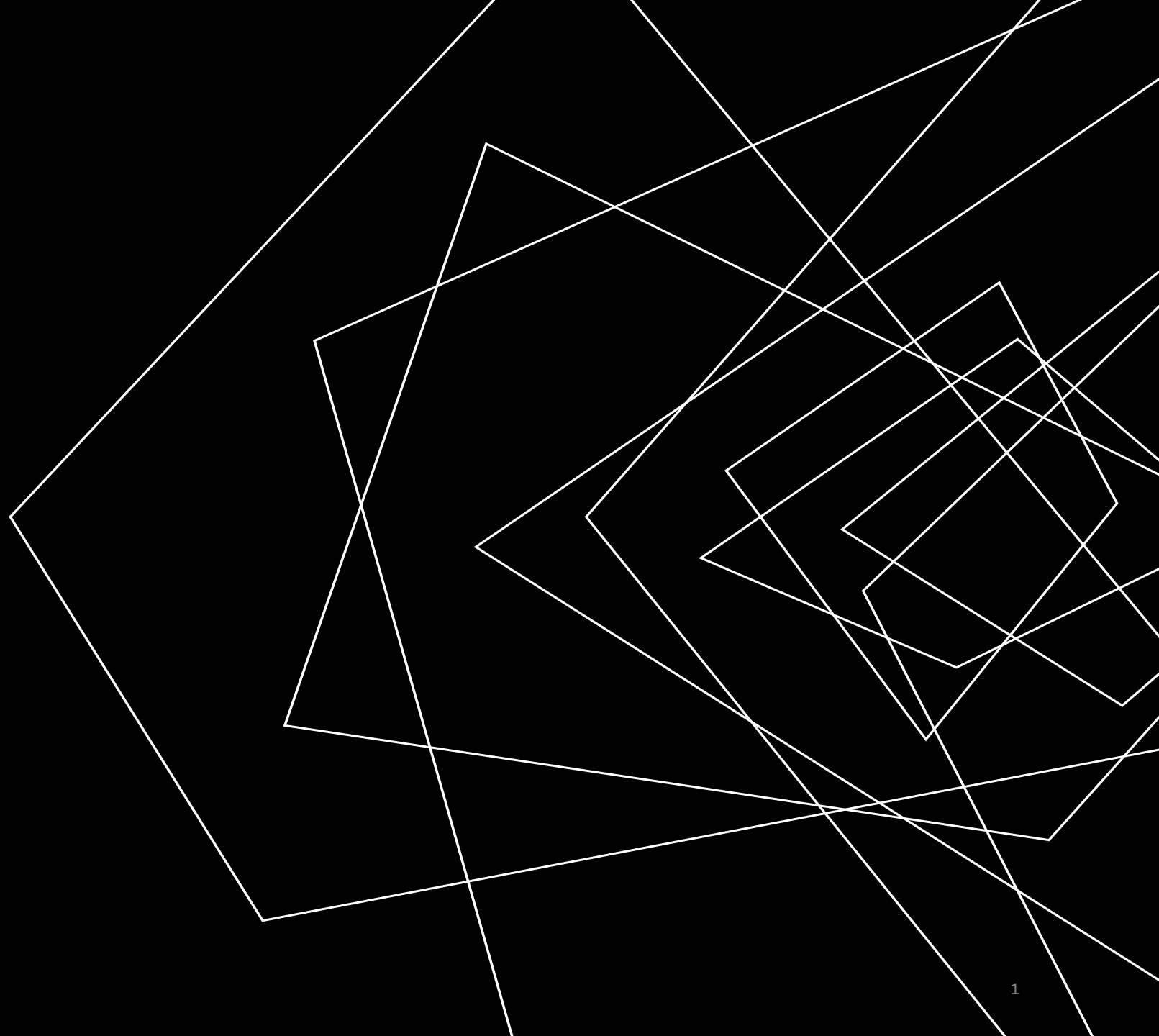


8-BIT WALLACE TREE MULTIPLIER

Shubham Saluja Kumar Agarwal
Purdue University



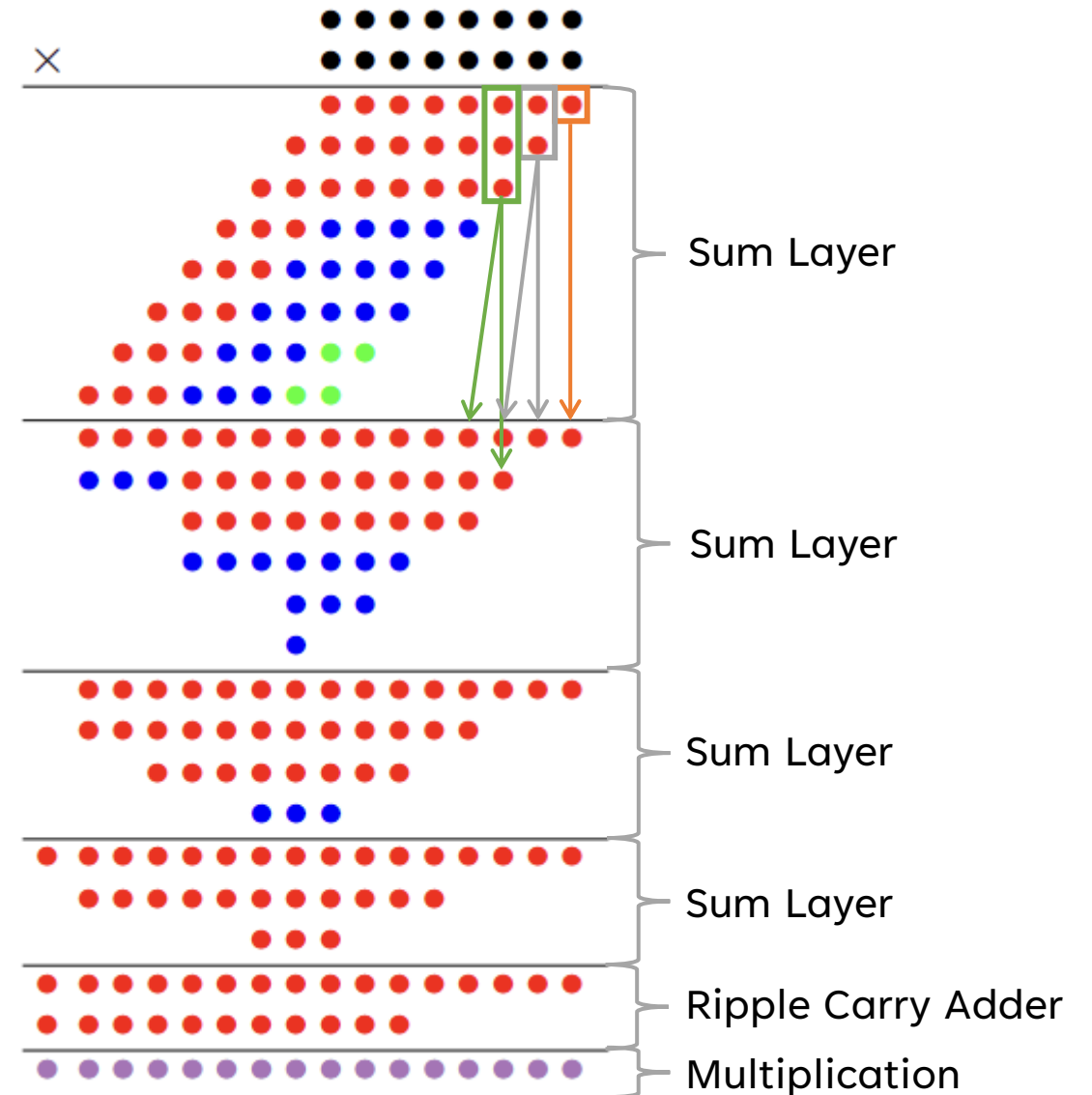
MULTIPLICATION PROCESS

- 1 bit: Buffer
- 2 bits: Half Adder
- 3 bits: Full Adder

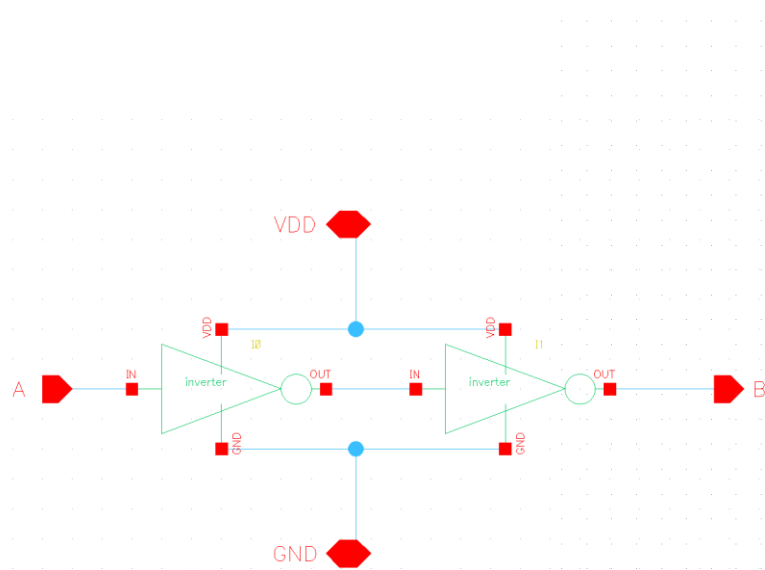
Adders send

- Sum to the same column
- Carry to the next column

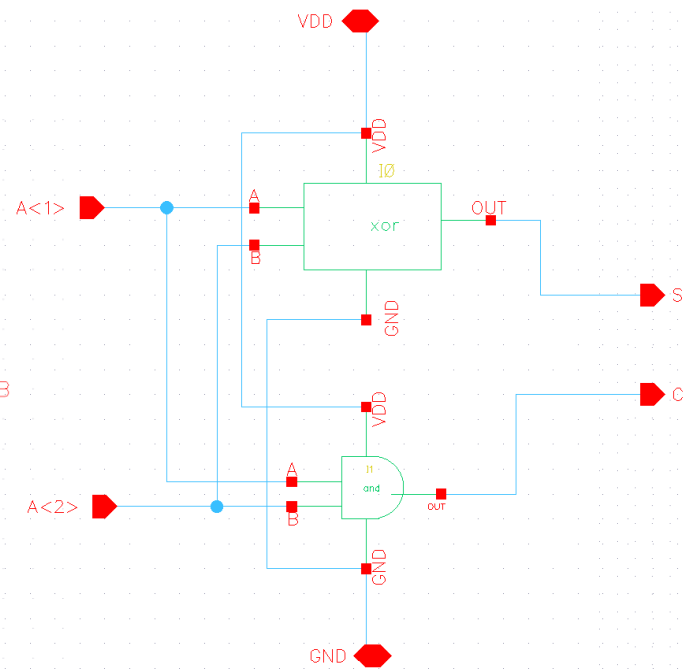
Ripple Carry Adder uses a full adder on any pair of bits, except the first pair, which uses a half adder



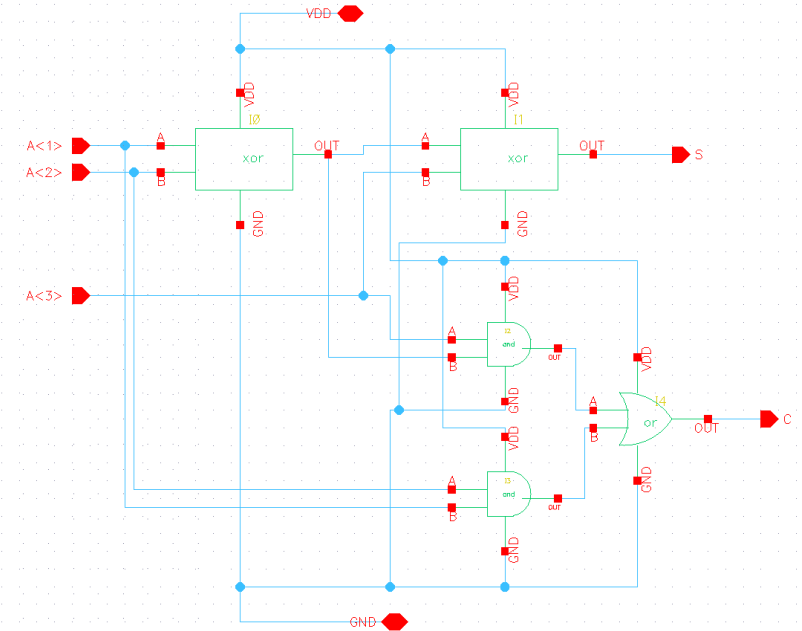
BUILDING BLOCK SCHEMATICS



Buffer

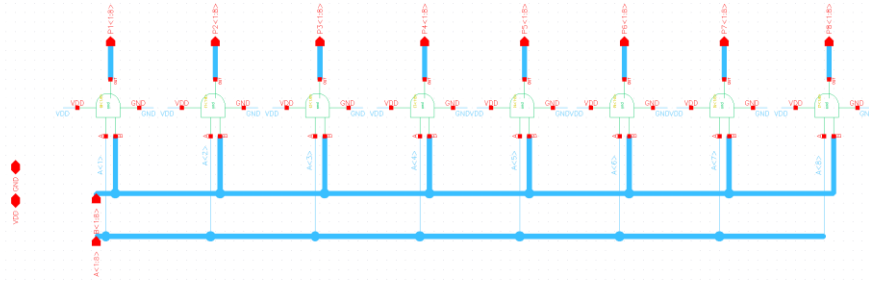


Half Adder

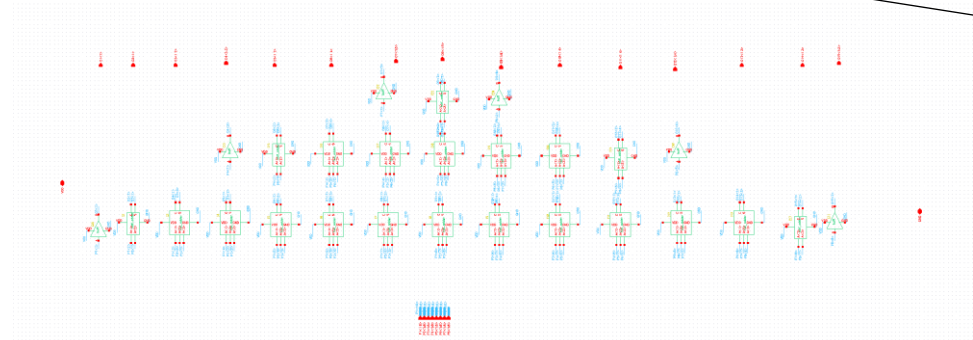


Full Adder

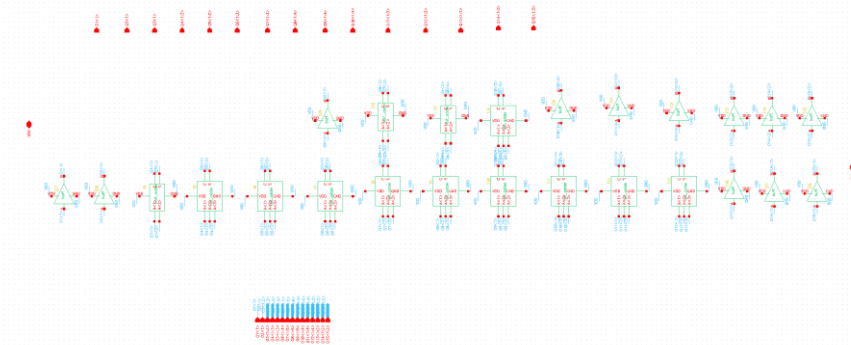
MODULE SCHEMATICS



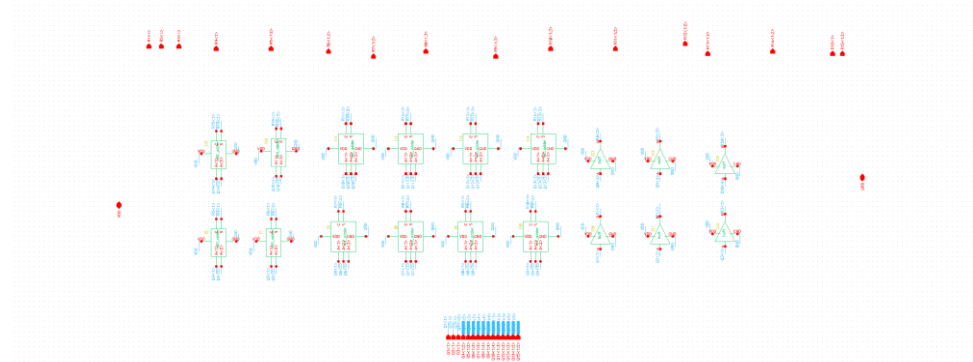
Partial Product Generator



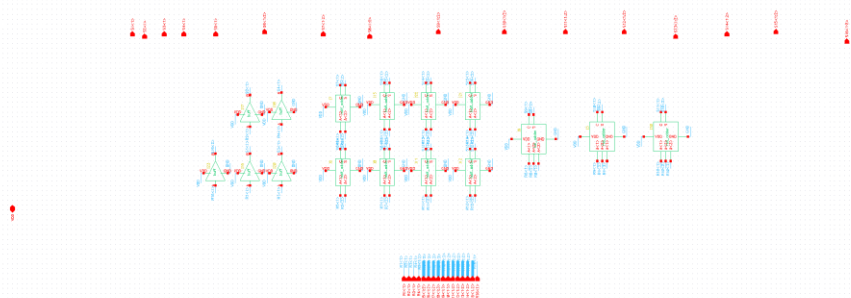
Sum Layer 1



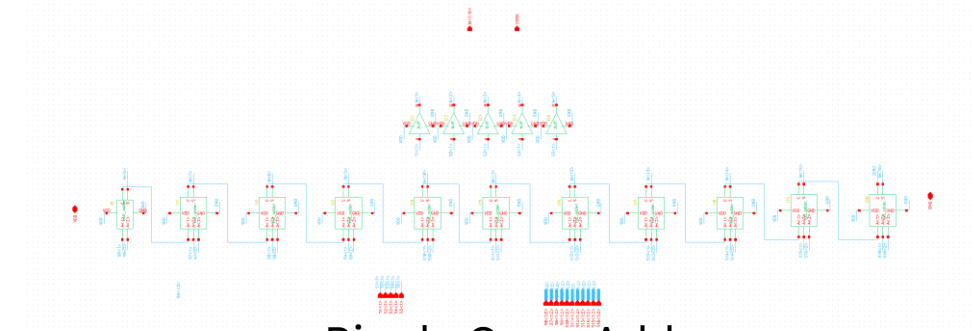
Sum Layer 2



Sum Layer 3

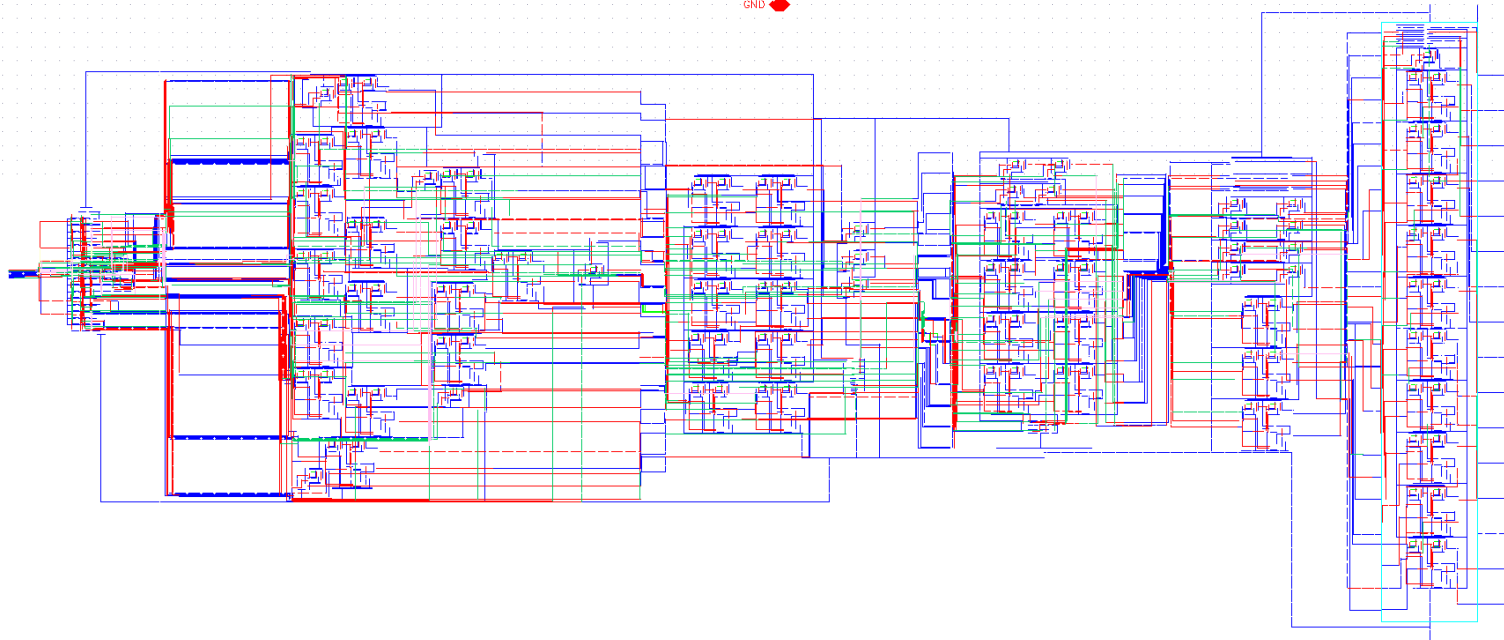
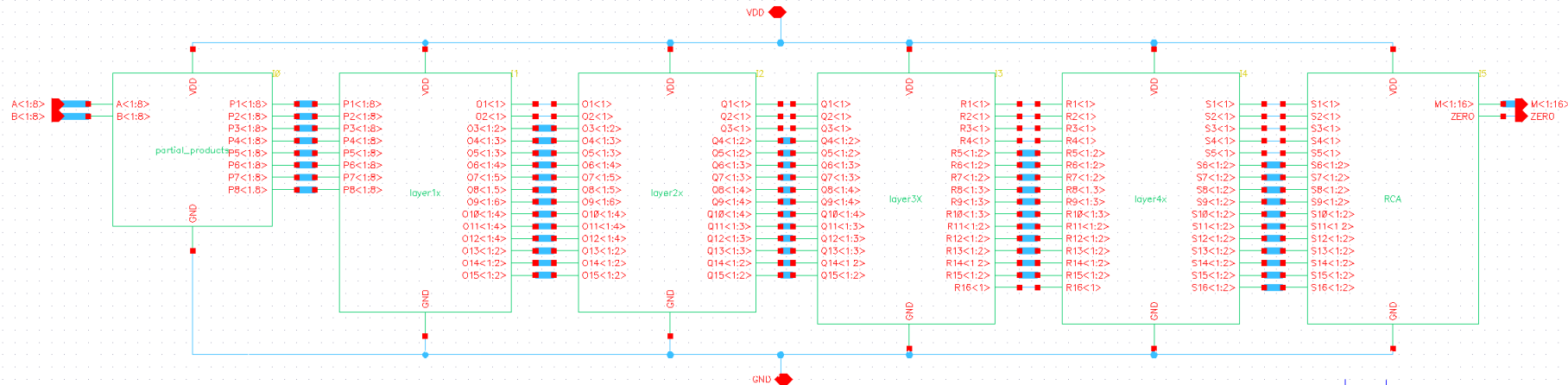


Sum Layer 4



Ripple Carry Adder

MULTIPLIER SCHEMATIC AND LAYOUT



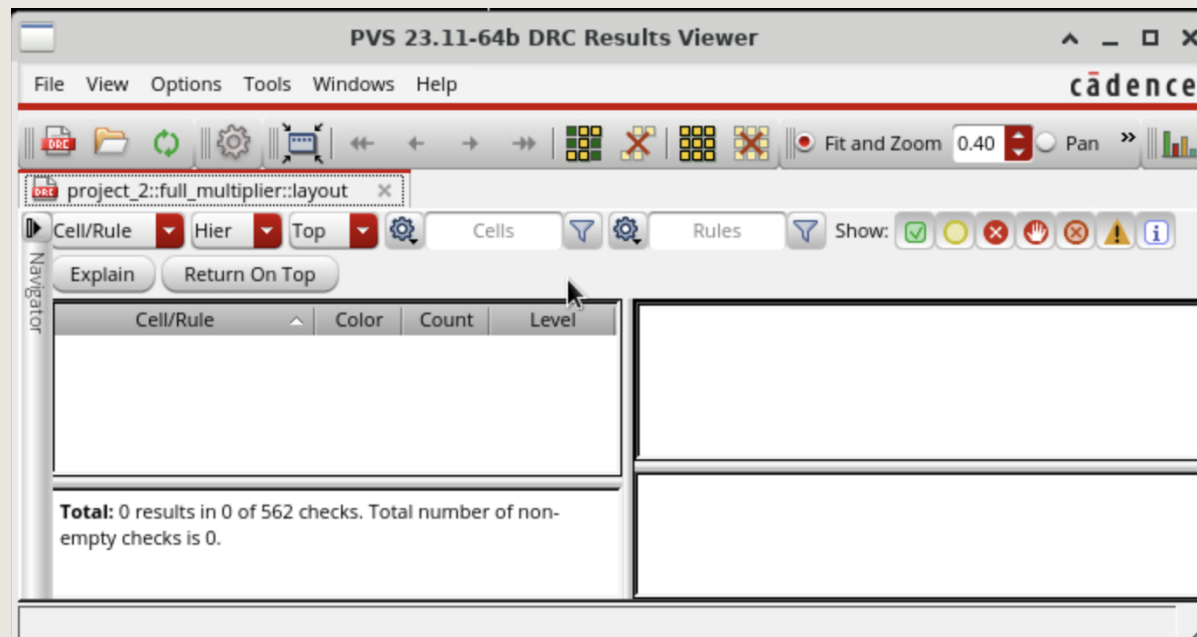
2474 transistors:

- 64 AND
- 40 FA
- 17 HA
- 26 Buffer

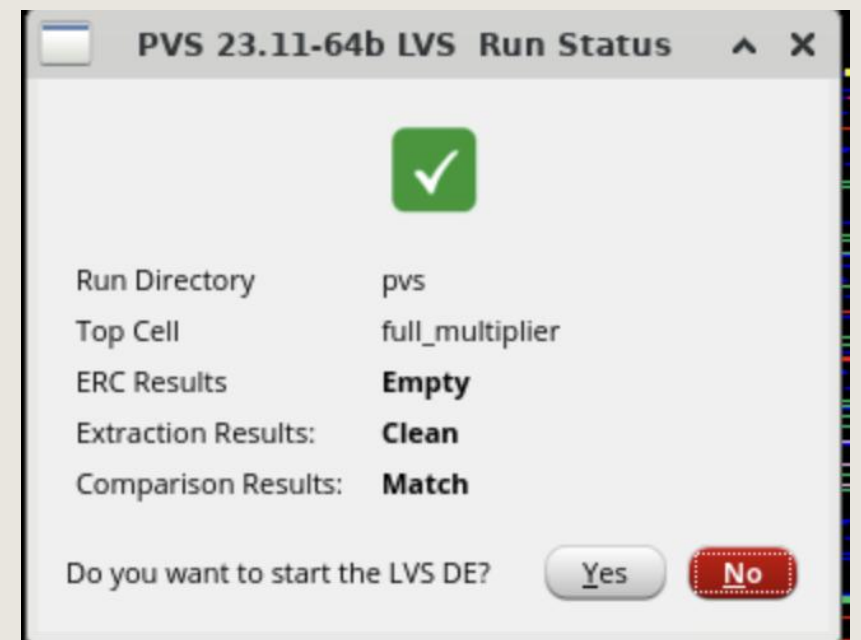
$\sim 0.15 \text{ mm}^2$

LAYOUT VERIFICATION

DRC verification



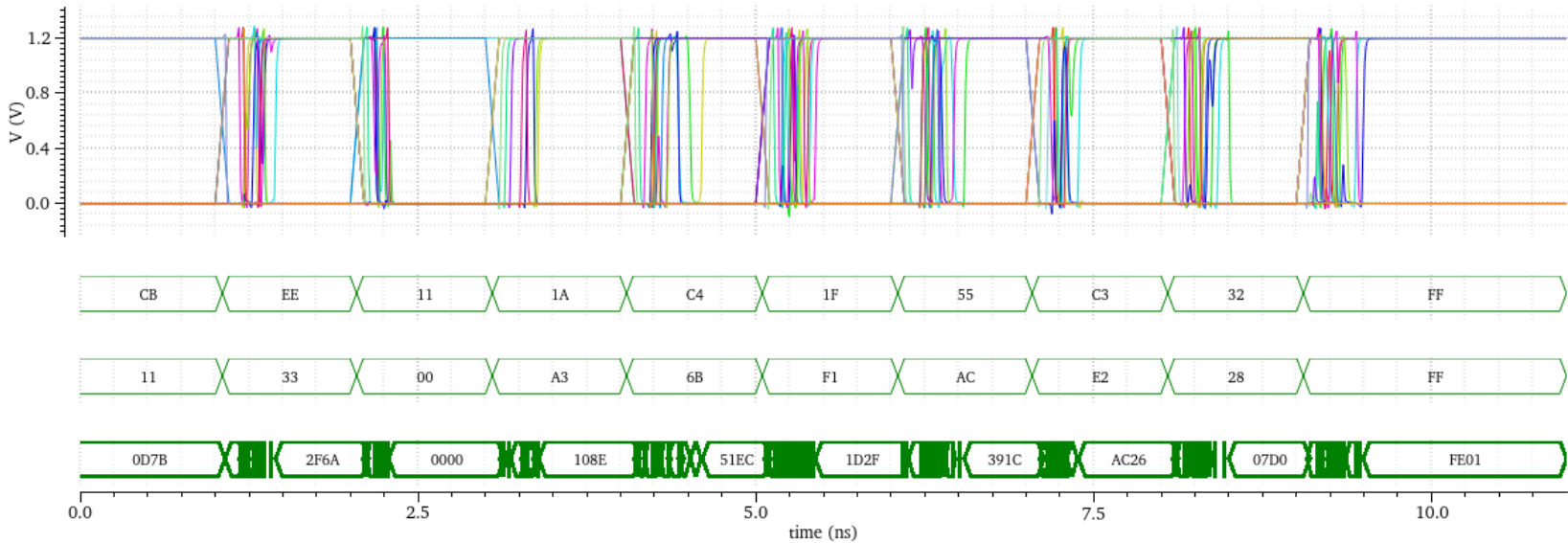
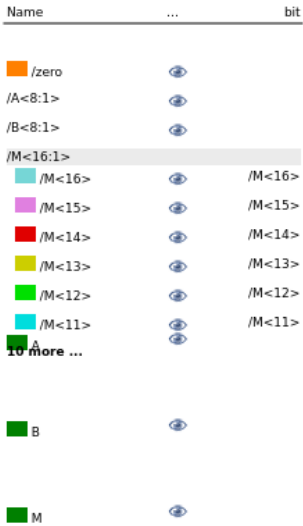
LVS verification



FUNCTIONALITY

A	B	M
CB	11	0D7B
EE	33	2F6A
11	00	0000
1A	A3	108E
C4	6B	51EC
1F	F1	1D2F
55	AC	391C
C3	E2	AC26
32	28	07D0
FF	FF	FE01

Transient Response



Thu Apr 24 18:34:02 20... 1

THEORETICAL ASSUMPTIONS

Minimum Sized Transistors

$$R_{NMOS} = 3k\Omega$$

$$R_{PMOS} = 6k\Omega$$

$$C_{unit} \approx 2fF$$

$$\alpha = 0.25$$

$$f = 1GHz \text{ (100MHz for post-layout)}$$

$$VDD = 1.2V$$

THEORETICAL ANALYSIS

Delay

Estimates for the delay are made using $t = 0.69RC$.

The total delay is the sum of the delay of each of the layers:

- Partial Products $\rightarrow t_{AND}$
- Sum Layers $\rightarrow 4 \times t_{FA,sum}$
- Ripple Carry Adder $\rightarrow 12 \times t_{FA,carry} + t_{FA,sum}$

Individual Delays:

- $t_{AND} = t_{NAND} + t_{INV} \approx 8.28ps + 4.14ps \approx \mathbf{12.5ps}$
- $t_{FA,sum} = 2 \times t_{XOR} \approx 2 \times (t_{INV} + 16.66ps) \approx \mathbf{40ps}$
- $t_{FA,carry} = t_{AND} + t_{OR} \approx 12.5ps + 18.75ps \approx \mathbf{31.25ps}$

Total Delay:

- $t_M \approx 5 \times 40ps + 12 \times 31.25ps + 12.5ps = \mathbf{587.5ps}$

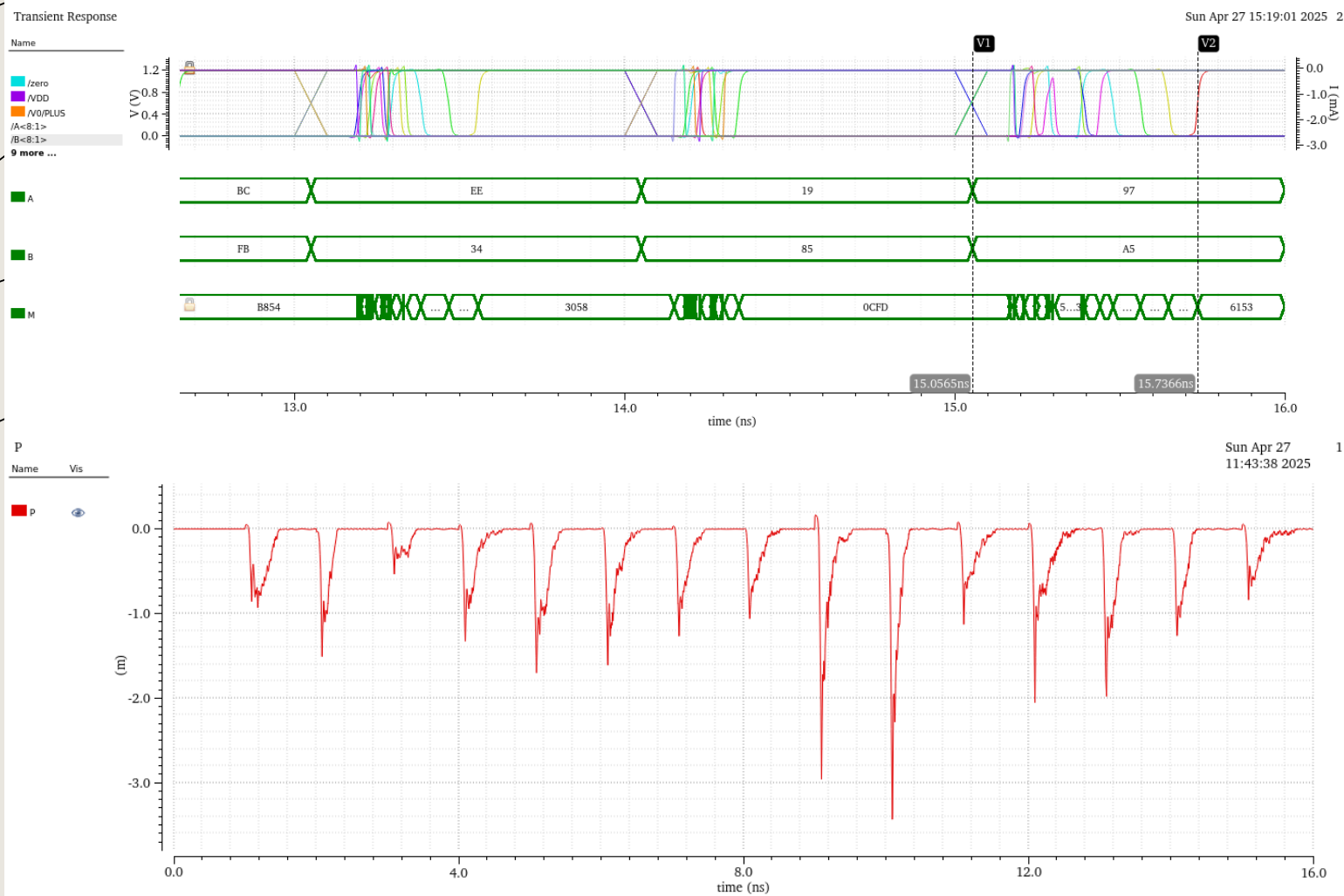
Power

- There were:
 - **40 full adders**
 - **17 half adders**
 - **26 buffers**
 - **64 AND gates**
- Approximated component capacitances:
 - $C_{FA} = 3C_{unit}$
 - $C_{HA} = 2C_{unit}$
 - $C_{AND} = C_{buffer} = C_{unit}$

$$\begin{aligned} C_{load} &= (40 \times 3C_{unit}) + (17 \times 2C_{unit}) + 26C_{unit} + 64C_{unit} \\ &= 244C_{unit} = 488fF \end{aligned}$$

$$\begin{aligned} P_{avg} &= \alpha \times C_{load} \times V_{dd}^2 \times f = \\ &0.25 \times 488fF \times (1.2V)^2 \times 1GHz = 175.68\mu W \end{aligned}$$

PRE-LAYOUT SIMULATION RESULTS



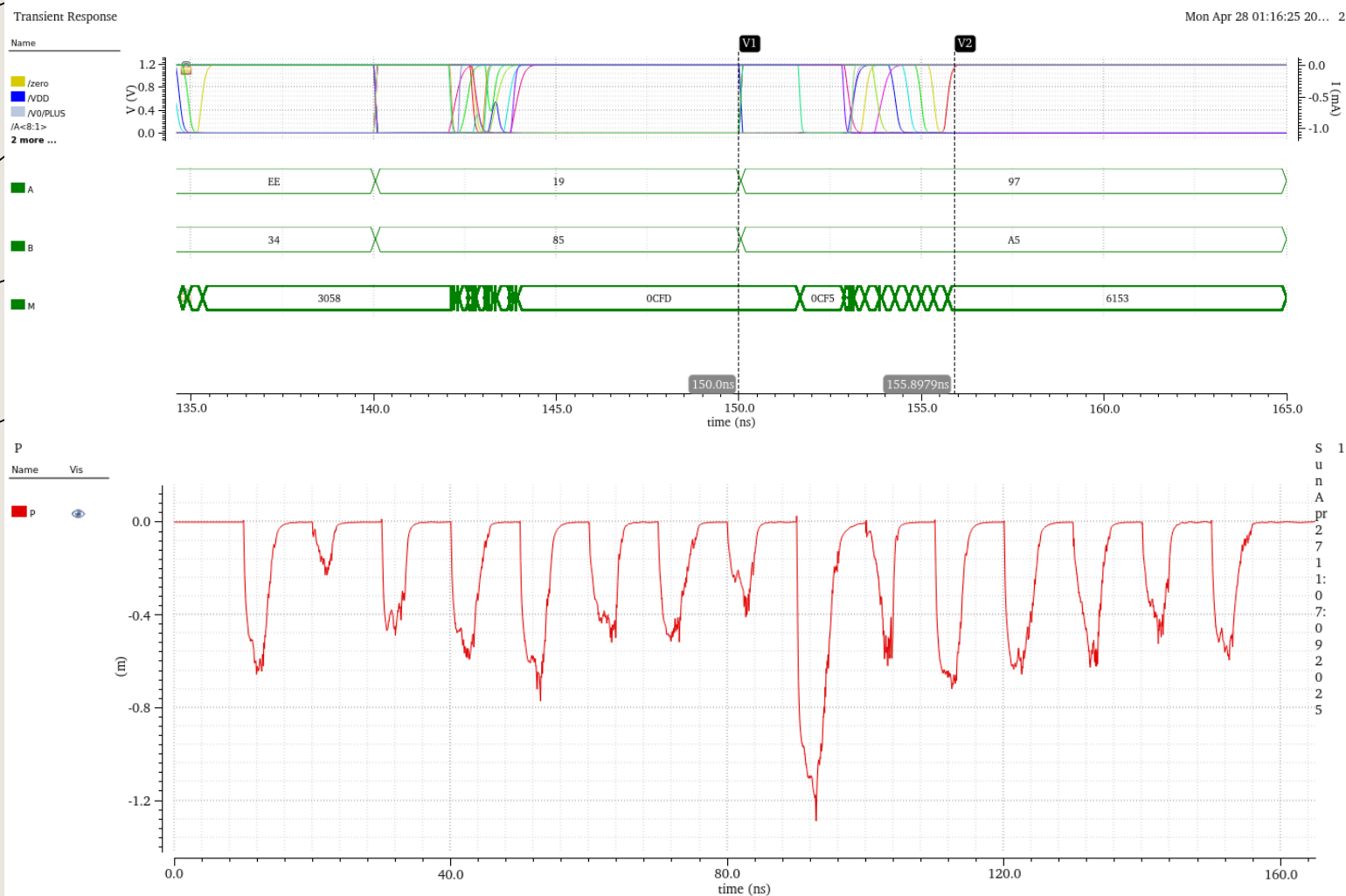
Delay

$$\Delta t = t_M = 680ps$$

Power

$$P_{avg} = avg(IV) = -186.2\mu W$$

POST-LAYOUT SIMULATION RESULTS



Delay

$$\Delta t = t_M = 5900ps$$

Power

$$P_{avg} = avg(IV) = -176.8\mu W$$

Note: This is run at 100MHz to avoid errors.

RESULT COMPARISONS

	Theoretical	Pre-Layout	Post-Layout
Propagation Delay	$587ps$	$680ps$	$5900ps$
Power (at 100MHz)	$17.568\mu W$	$18.23\mu W$	$176.8\mu W$